Coulomb’s law

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Learning target: Relate electric force to charge magnitude and separation.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 17.*
The farther target takes 3 times as long because t=d/v.

02. Worked example: explain the model
Equal opposite repulsive forces; larger separation yields weaker force by factor 4.

03. Read and interpret the model
Attraction remains; each force becomes one fourth.

04. Apply the physics
3 N.

05. Explain your reasoning
No; the interaction forces have equal magnitudes by Newton’s third law.

06. Review: connect an earlier idea *Review: Unit 6, Day 8.*
Parallel oscillation; a longitudinal pressure wave.

07. Exit check
It triples.

08. STRETCH: test the model
12×2/4=6 N; sign relationship unchanged unless a charge sign changes.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 17.*
v≈17.32 m/s; t≈288.7 s. Requires the long-wave shallow-water regime and simplified depth; not a damage forecast.

02. Worked example: explain the model
Equal opposite repulsive forces; larger separation yields weaker force by factor 4.

03. Read and interpret the model
Attraction remains; each force becomes one fourth.

04. Apply the physics
0.60 N; attraction.

05. Explain your reasoning
No; the interaction forces have equal magnitudes by Newton’s third law.

06. Review: connect an earlier idea *Review: Unit 6, Day 8.*
240 cycles; T≈0.001667 s.

07. Exit check
4/9 of the original force.

08. STRETCH: test the model
Initial force = $9 \times 10^{9} \times 4 \times 10^{-12} / 0.20^{2} = 0.90$ N, repulsive. At 0.60 m, force = 0.90/9 = 0.10 N, still repulsive.